Ex No.:3 Poorna Sri.A
Date: 953625148038

1. Aim:

To write a C++ program to reverse a given string.

Algorithm:

Step 1: Start the program.
Step 2: Read a string from the user.
Step 3: Find the length of the string.
Step 4: Initialize two indices, one at the beginning and one at the end of the string.
Step 5: Swap the characters at the beginning and end indices.
Step 6: Move the beginning index forward and the end index backward.
Step 7: Repeat Steps 5 and 6 until the indices meet or cross each other.
Step 8: Display the reversed string.
Step 9: Stop the program.

Program Code:

```
#include<iostream>
#include<string>
using namespace std;
int main(){
   string str;
   cout << "Enter a string: ";
   getline(cin, str);
   int n = str.length();
   for (int i = 0; i < n / 2; i++){
      char temp = str[i];
      str[i] = str[n - i - 1];
      str[n - i - 1] = temp;
   }
   cout << "Reversed String: " << str << endl;
   return 0;
}
```

Sample Output:

```
Z:\DS\Code\EX-3>
Z:\DS\Code\EX-3>g++ reverse.cpp -o reverse.exe

Z:\DS\Code\EX-3>reverse.exe
Enter a string: Hello
Reversed String: olleH
```

Result:

Thus, the C++ program to reverse a given string was executed successfully and the reversed string was displayed.

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2. Aim:

To write a C++ program to reverse each word in a given sentence while maintaining the original word order.

Algorithm:

Step 1: Start the program.
Step 2: Read a string from the user.
Step 3: Find the length of the string.
Step 4: Initialize two indices, one at the beginning and one at the end of the string.
Step 5: Swap the characters at the beginning and end indices.
Step 6: Move the beginning index forward and the end index backward.
Step 7: Repeat Steps 5 and 6 until the indices meet or cross each other.
Step 8: Display the reversed string.
Step 9: Stop the program.

Program code:

```
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;
int main() {
   string str, word;
   cout << "Enter a sentence: ";
   getline(cin, str);
   for (int i = 0; i <= str.length(); i++) {
      if (i == str.length() || str[i] == ' ') {
         reverse(word.begin(), word.end());
         cout << word << " ";
         word = "";
      } else {
         word += str[i];
      }
   }
   return 0;
}
```

Sample Output:

```
Z:\DS\Code\EX-3>g++ revwordsen.cpp -o revwordsen.exe

Z:\DS\Code\EX-3>revwordsen.exe
Enter a sentence: I am Nithya
I ma ayhtiN
```

Result:

Thus, the C++ program to reverse each word in a sentence was executed successfully and the output was displayed.

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3. Aim:

To write a C++ program to evaluate a postfix expression using a stack.

Algorithm:

Step 1: Start the program.
Step 2: Read the postfix expression from the user.
Step 3: Create an empty stack.
Step 4: Scan the expression from left to right.
Step 5: If the current symbol is an operand, push it onto the stack.
Step 6: If the current symbol is an operator, pop the top two operands from the stack.
Step 7: Perform the required arithmetic operation.
Step 8: Push the result back onto the stack.
Step 9: Repeat Steps 5 to 8 until the entire expression is processed.
Step 10: Display the element at the top of the stack as the final result.
Step 11: Stop the program.

Program Code:

```
#include<iostream>
#include<stack>
#include<string>
using namespace std;
int main() {
    string exp;
    cout << "Enter Postfix Expression: ";
    cin >> exp;
    stack<int> s;
    for (char ch : exp) {
        if (isdigit(ch)) {
            s.push(ch - '0');
        } else {
            int val2 = s.top();
            s.pop();
            int val1 = s.top();
            s.pop();
            switch (ch) {
                case '+':
                    s.push(val1 + val2);
                    break;
                case '-':
                    s.push(val1 - val2);
                    break;
                case '*':
                    s.push(val1 * val2);
```

```
                    break;
                case '/':
                    s.push(val1 / val2);
                    break;
            }
        }
    }
    cout << "Result = " << s.top() << endl;
    return 0;
}
```

Sample Output:

```
Z:\DS\Code\EX-3>g++ postfix.cpp -o postfix.exe

Z:\DS\Code\EX-3>postfix.exe
Enter Postfix Expression: 23+
Result = 5
```

Result:

Thus, the C++ program to evaluate a postfix expression using a stack was executed successfully and the correct result was displayed.